

A.2 Forces and Momentum

Past Paper Answers

These are past paper answers selected from past papers (2016 - 2020) that relate to the topic A.2 Forces and Momentum.

The associated questions are in the separate accompanying document

Answer 1

D

Answer 2

C

Answer 3

A

Answer 4

no force/friction on the block, hence constant motion/velocity/speed ✓		1
force acts on block OR gravity/component of weight pulls down slope ✓ velocity/speed decreases OR it is slowing down OR it decelerates ✓	Do not allow a bald statement of "N2" or " $F = ma$ " for MP1. Treat references to energy as neutral.	2

Answer 5

C

Answer 6

C

Answer 7

B

Answer 8

D

Answer 9

a		determines component correctly / $15 \sin 50$ seen ✓ $s = \frac{(15 \sin 50)^2}{2 \times 9.81} = 6.7 \text{ «m»} \checkmark$ correct reasoning consistent with candidate data ✓	Allow method via $v = u + at$. Allow use of $g = 10 \text{ m s}^{-2}$, gives 6.6 m and 8.1 m. Allow [2 max] for use of $15 \cos 50$, gives 4.7 m and 6.2 m. Allow [1 max] (as MP2) if 13 m is obtained due to use of 15 m s^{-1} rather than $15 \sin$ or $15 \cos 50$. If no unit given, assume metre.	3
b	i	$F = \frac{(0.058 \times 15)}{0.023} = 38 \text{ «N» OR } 37.8 \text{ «N»} \checkmark$	Do not penalise sf here. Working not required.	1
b	ii	force of ball on racket is equal to force of racket on ball or is 38 N ✓ ball exerts force in opposite direction to force of racket on ball ✓	Do not accept "same force". Allow ECF from force value in bi Accept "opposite force" for "in opposite direction". Do not accept undefined references to "reaction" the direction of the forces must be clear.	2

Answer 10

a		evidence that area under graph used OR use of mean velocity $\times$ time ✓ $\frac{29.8 \times 2}{17.5} = 3.41 \text{ «ms}^{-1}\text{»} \checkmark$	Award [2] for a bald correct answer. Award [1] for 1.70 ms^{-1} .	2
b		ALTERNATIVE 1 $\text{«deceleration»} = \frac{3.41}{14.0} = 0.243 \text{ ms}^{-2} \checkmark$ $F = 0.243 \times m \checkmark$ $\mu = \frac{0.243 \times m}{m \times 9.81} = 0.025 \checkmark$ ALTERNATIVE 2 distance travelled after release = $23.85 \text{ «m»} \checkmark$ KE lost = $5.81m \text{ «J»} \checkmark$ $\mu_d = \frac{\text{KE lost}}{mg \times \text{distance}} = \frac{5.81m}{23.85mg} = 0.025 \checkmark$	Award [3] for a bald correct answer. Ignore sign in acceleration. Allow ECF from (a) (note that $\mu = 0.0073 \times$ candidate answer to (a)). Ignore any units in answer. Condone omission of m in solution. Allow $g = 10 \text{ N kg}^{-1}$ (gives 0.024).	3

c	normal force, upwards, ignore point of application ✓ weight/weight force/force of gravity, downwards, ignore point of application ✓ friction/resistive force, to left, at bottom of stone, point of application must be on the interface between ice and stone ✓	<i>Force must be labeled for its mark to be awarded. Blob at poa not required.</i> <i>Allow OWTTE for normal force. Allow N, R, reaction.</i> <i>The vertical forces must lie within the middle third of the stone.</i> <i>Allow mg, W but not "gravity".</i> <i>Penalise gross deviations from vertical/horizontal once only.</i> <i>Allow F, μR. Only allow arrows/lines that lie on the interface. Take the tail of the arrow as the definitive point of application and expect line to be drawn horizontal.</i> <i>Award [2 max] if any force arrow does not touch the stone</i> <i>Do not award MP3 if a "driving force" is shown acting to the right. This need not be labelled to disqualify the mark.</i> <i>Treat arrows labelled "air resistance" as neutral.</i> N.B: <i>Diagram in MS is drawn with the vertical forces not collinear for clarity.</i>	3
---	--	---	---

Answer 11

B

Answer 12

A

Answer 13

B

Answer 14

i	«76×9.6» = 730 ✓ Ns OR kg ms^{-1} ✓		2
ii	safety net extends stopping time ✓ $F = \frac{\Delta p}{\Delta t}$ therefore F is smaller «with safety net» OR force is proportional to rate of change of momentum therefore F is smaller «with safety net» ✓	Accept reverse argument.	2

Answer 15

$\omega = \frac{v}{r} = \frac{27}{0.6} = 45 \checkmark$ $\text{rad s}^{-1} \checkmark$	<i>Do not accept Hz.</i> <i>Award [1 max] if unit is missing.</i>	2
---	--	----------

Answer 16

D

Answer 17

A

Answer 18

a		arrow vertically downwards labelled weight «of sledge and/or girl» $W/mg/\text{gravitational force}/F_g/F_{\text{gravitational}}$ AND arrow perpendicular to the snow slope labelled reaction force $R/\text{normal contact force}/N/F_N$ ✓ friction force F/f acting up slope «perpendicular to reaction force» ✓	Do not allow G/g "gravity". Do not award MP1 if a "driving force" is included. Allow components of weight if correctly labelled. Ignore point of application or shape of object. Ignore "air resistance". Ignore any reference to "push of feet on sledge". Do not award MP2 for forces on sledge on horizontal ground The arrows should contact the object	2
b		gravitational force/weight from the Earth «downwards» ✓ reaction force from the sledge/snow/ground «upwards» ✓ no vertical acceleration/remains in contact with the ground/does not move vertically as there is no resultant vertical force ✓	Allow naming of forces as in (a) Allow vertical forces are balanced/equal in magnitude/cancel out	3
c		mention of conservation of momentum OR $5.5 \times 4.2 = (55 + 5.5) \llcorner v \llcorner$ ✓ $0.38 \llcorner \text{m s}^{-1} \llcorner$ ✓	Allow $p = p'$ or other algebraically equivalent statement Award [0] for answers based on energy	2
d		same change in momentum/impulse ✓ the time taken «to stop» would be greater «with the snow» ✓ $F = \frac{\Delta p}{\Delta t}$ therefore F is smaller «with the snow» OR force is proportional to rate of change of momentum therefore F is smaller «with the snow» ✓	Allow reverse argument for ice	3
e	i	«friction force down slope» $= \mu mg \cos(6.5) = \llcorner 5.9 \text{N} \llcorner$ ✓ «component of weight down slope» $= mg \sin(6.5) \llcorner = 6.1 \text{N} \llcorner$ ✓ «so $a = \frac{F}{m}$ » acceleration $= \frac{12}{5.5} = 2.2 \llcorner \text{m s}^{-2} \llcorner$ ✓	Ignore negative signs Allow use of $g = 10 \text{ m s}^{-2}$	3
e	ii	correct use of kinematics equation ✓ distance $= 4.4$ or $4.0 \llcorner \text{m} \llcorner$ ✓	Allow ECF from (e)(i)	2
f		calculates a maximum value for the frictional force $= \llcorner \mu R = \llcorner 7.5 \llcorner \text{N} \llcorner$ ✓ sledge will not move as the maximum static friction force is greater than the component of weight down the slope ✓	Allow correct conclusion from incorrect MP1 Allow $7.5 > 6.1$ so will not move	2

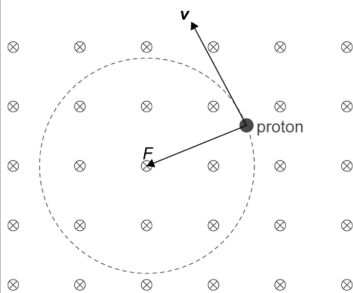
Answer 19

B

Answer 20

D

Answer 21

a	i	F towards centre ✓		1
a	ii	v tangent to circle and in the direction shown in the diagram ✓	 The diagram shows a grid of 'x' marks representing a magnetic field directed into the page. A dashed circle represents the path of a proton. A solid dot on the circle is labeled 'proton'. A vector F points from the proton towards the center of the circle. A vector v is tangent to the circle at the proton's position, pointing in the counter-clockwise direction.	1